MSK Physiology — NBME Study Guide

Neuromuscular junction · Skeletal muscle E–C coupling · Contraction mechanics · Energetics & fiber types · Exercise · Smooth muscle

Red = extra high yield (classic NBME) Black = core content *Italic blue = mnemonic*

1. Neuromuscular junction (NMJ)

1. AP reaches motor nerve terminal → **voltage-gated Ca^{2+} channels (P/Q type) open → Ca^{2+} influx**.
2. Ca^{2+} triggers **SNARE**-mediated exocytosis of ACh vesicles (synaptobrevin, syntaxin, SNAP-25).
3. ACh binds **nicotinic receptors (N_M, ligand-gated cation channels; Na^+ in > K^+ out)** on the motor end plate (junctional folds).
4. End-plate potential (graded) → always reaches threshold (high safety factor) → muscle AP.
5. **Acetylcholinesterase** in the synaptic cleft hydrolyzes ACh → acetate + choline (choline recycled).

Agent / disease	Mechanism	Effect
Botulinum toxin	**Cleaves SNAREs → blocks ACh release**	**Descending flaccid paralysis; infant botulism (honey); cosmetic/dystonia use**
Myasthenia gravis	**Antibodies vs postsynaptic nicotinic ACh receptors**	**Fatigable weakness, ptosis, diplopia, worse with use; thymoma; edrophonium/AChE inhibitor test; treat pyridostigmine**
Lambert-Eaton	**Antibodies vs presynaptic voltage-gated Ca^{2+} channels**	**Proximal weakness improves with repeated use, autonomic sx, ↓ reflexes; small cell lung cancer**
Curare / d-tubocurarine, rocuronium	Competitive nicotinic antagonists (nondepolarizing)	Paralysis; reversed by AChE inhibitors (neostigmine) / sugammadex
Succinylcholine	**Depolarizing agonist; persistent depolarization inactivates Na^+ channels**	**Fasciculations then flaccid paralysis; malignant hyperthermia, hyperkalemia**
Organophosphates / neostigmine	**Inhibit AChE**	**↑ ACh; organophosphate poisoning = DUMBBELSS; atropine + pralidoxime**
Hemicholinium	Blocks choline reuptake	↓ ACh synthesis
α-bungarotoxin	Irreversibly binds nicotinic receptor	Paralysis

2. Skeletal muscle structure & excitation–contraction coupling

- Sarcomere (Z to Z): **A band constant; I band and H zone shorten** (sliding filament). Titin = elasticity/centering; nebulin = actin length ruler.
- **E–C coupling (skeletal): muscle AP spreads down T-tubules → DHPR (L-type Ca^{2+} channel acting as a voltage sensor) is mechanically coupled to RyR1 on the SR → Ca^{2+} release → binds troponin C → tropomyosin moves off myosin-binding sites → cross-bridge cycling.** Extracellular Ca^{2+} is NOT needed in skeletal muscle.
- **Cardiac**: Ca^{2+}-induced Ca^{2+} release — Ca^{2+} entering through DHPR triggers RyR2 (extracellular Ca^{2+} required).
- **Relaxation: SERCA pumps Ca^{2+} back into the SR (uses ATP)**; phospholamban regulates SERCA in heart. Calsequestrin stores Ca^{2+} in SR.
- **Malignant hyperthermia**: RyR1 mutation (AD); triggered by volatile anesthetics/succinylcholine → uncontrolled Ca^{2+} release → rigidity, hyperthermia, hypercarbia, hyperkalemia, ↑ CK → **dantrolene** (blocks RyR1).

Cross-bridge cycle (ATP roles): (1) Rigor: myosin bound to actin, no ATP. (2) **ATP binds myosin → myosin releases actin.** (3) ATP hydrolysis (ADP + Pi stay bound) → myosin head "cocked". (4) Head binds actin (once Ca^{2+} exposes sites). (5) **Pi release → power stroke**; ADP released. **No ATP → rigor mortis (heads stuck to actin).** ATP is also used by SERCA (relaxation) and Na^+/K^+-ATPase.

3. Contraction mechanics

- **Twitch**: single AP → single contraction (latent, contraction, relaxation). AP lasts ~2–5 ms, twitch much longer → summation possible.
- **Summation & tetanus**: ↑ stimulation frequency → Ca^{2+} stays elevated → summation → **fused tetanus** (max force). **Treppe** (staircase): slightly ↑ force with repeated twitches.
- **Recruitment (size principle)**: small motor units (type I, slow, fatigue-resistant) recruited first, large (type II) last. Force graded by recruitment + rate coding.
- Motor unit = one α motor neuron + all fibers it innervates; small units for fine control (eye muscles), large for gross (quadriceps).
- **Isometric** (length constant, tension ↑) vs **isotonic** (constant load, length changes): concentric (shortening), **eccentric** (lengthening under load — most force, most DOMS/damage).
- **Length–tension relationship**: max active tension at optimal sarcomere length (~2.0–2.2 µm, maximal cross-bridge overlap). Too short (filaments overlap/collide) or too long (little overlap) → ↓ active tension. Passive tension (titin, CT) ↑ with stretch; total = active + passive.
- **Force–velocity relationship**: ↑ load → ↓ shortening velocity; Vmax at zero load; at max (isometric) load velocity = 0. Power max at ~1/3 Vmax.
- Preload = load before contraction (sets length); afterload = load the muscle works against.

4. Energetics & fiber types

- ATP stores last only seconds. Sources: **phosphocreatine (creatine kinase) — fastest, ~10 s** → anaerobic glycolysis (→ lactate; ~1–2 min) → oxidative phosphorylation (sustained, needs O_2).
- **Fenn effect**: muscle doing more work consumes more ATP/energy (energy liberated ∝ work performed).
- Efficiency ~20–25% (rest lost as heat).
- Fatigue: ↓ glycogen, ↑ Pi, ↑ H^+, impaired Ca^{2+} release; central fatigue.

	Type I (slow oxidative)	Type IIa (fast oxidative-glycolytic)	Type IIx/IIb (fast glycolytic)
Color	Red (myoglobin)	Pink/red	White
Mitochondria / capillaries	Many	Many	Few
Myosin ATPase speed	Slow	Fast	Fastest
Fatigue	Resistant	Intermediate	Fast
Fiber size / motor unit	Small	Medium	Large
Use	Posture (soleus), endurance	—	Sprint, power, extraocular-like fast movement

"One slow red ox": type 1, slow, red, oxidative. Endurance training ↑ mitochondria/type I traits; resistance training → hypertrophy of type II.

5. Exercise physiology

- ↑ Cardiac output (HR and SV), ↓ TPR (local metabolic vasodilation: adenosine, K^+, H^+, CO_2, lactate, NO), ↑ systolic BP, diastolic BP ~same or ↓.
- Ventilation ↑ proportionally (PaO_2, $PaCO_2$ normal in moderate exercise); ↑ O_2 extraction (rightward shift of O_2–Hb curve: ↑ temperature, CO_2, H^+, 2,3-BPG).
- **Oxygen debt (EPOC)**: extra O_2 after exercise to restore phosphocreatine, ATP, lactate clearance (Cori cycle), myoglobin O_2.
- Lactate threshold: point where lactate accumulates (anaerobic glycolysis predominates).
- Training: ↑ VO_2max, ↑ stroke volume, ↓ resting HR, ↑ capillary density, ↑ mitochondria.
- Rhabdomyolysis (crush, extreme exertion, statins): ↑ CK, myoglobinuria (urine + blood on dipstick, no RBCs) → AKI; hyperkalemia.

6. Smooth muscle

- No sarcomeres/troponin; actin anchored to **dense bodies** (α-actinin) and membrane; intermediate filaments (desmin, vimentin). Little SR; caveolae.
- **Contraction: ↑ intracellular Ca^{2+} (from ECF via L-type channels, and SR via IP_3 receptors) → Ca^{2+}–calmodulin → activates myosin light-chain kinase (MLCK) → phosphorylates myosin light chain → cross-bridge cycling.**
- **Relaxation: myosin light-chain phosphatase (MLCP) dephosphorylates myosin.** NO → ↑ cGMP → PKG → activates MLCP → relaxation (nitrates, sildenafil). β_2 / cAMP → PKA → inhibits MLCK → relaxation (albuterol).
- **Latch state**: dephosphorylated myosin remains attached → sustained tension with little ATP (sphincters, vascular tone).
- Ca^{2+} sources: voltage-gated (L-type; blocked by verapamil, nifedipine) = electromechanical; **pharmacomechanical** = receptor (α_1, M_3) → Gq → PLC → IP_3 → SR Ca^{2+} release (no AP needed). Rho kinase inhibits MLCP → Ca^{2+} sensitization.
- **Single-unit (visceral)**: gap junctions, contract as a syncytium, pacemaker/slow waves, stretch-activated — GI tract, uterus, bladder, small vessels. **Multi-unit**: each fiber innervated separately, fine control — iris, ciliary muscle, vas deferens, piloerector, large arteries.
- Slow waves (GI, interstitial cells of Cajal) set rhythm; spike potentials (Ca^{2+}) cause contraction.
- Slower, more economical, can contract over a wide length range (stress–relaxation/plasticity — bladder fills without pressure rise).

	Skeletal	Smooth
Ca^{2+} sensor	**Troponin C**	**Calmodulin → MLCK**
Ca^{2+} source	SR only (mechanical DHPR–RyR)	ECF + SR (IP_3)
Regulation	Thin-filament	Thick-filament (myosin phosphorylation)
Speed / ATP use	Fast / high	Slow / low (latch)

7. Rapid-fire NBME pearls

- **Myasthenia = postsynaptic AChR Ab (worse with use); Lambert–Eaton = presynaptic Ca^{2+} channel Ab (better with use, SCLC).**
- **Botulinum blocks ACh release (SNAREs); organophosphates block AChE; succinylcholine depolarizing block.**
- **Skeletal E–C: DHPR mechanically opens RyR1; cardiac: Ca^{2+}-induced Ca^{2+} release.**
- **ATP binding releases myosin from actin; no ATP = rigor. SERCA relaxes muscle.**
- **Malignant hyperthermia: RyR1, volatile anesthetics/succinylcholine → dantrolene.**
- **Smooth muscle: Ca^{2+}–calmodulin–MLCK; relax via MLCP (NO/cGMP); latch state.**
- **Size principle: small type I units first. A band constant.**